QIWEI MA

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EDUCATION

Shenzhen University (SZU) *June 2026*
M.S. in Artificial Intelligence
GPA: 3.5/4.0

Ningxia University (NXU) *June 2022*
B.E. in Materials
GPA: 3.0/4.0

INTERESTS

- Robotics and Embodied Intelligence
- AI in Education
- Large Language Models and Multi-Agent Systems

RESEARCH EXPERIENCE

Multi-Agent Conversational AI for EFL Speaking Practice [1] Jan 2025 - Jan 2026
Supervisor: Dr. Zhang SZU

- Developed a multi-agent system (MAS) for EFL speaking practice.
- Built seven specialized agents for preprocessing, response generation, and dialogue supervision.
- Investigated the mechanisms underlying MAS superiority and identified synergistic effects among integrated features.
- Conducted a 4-week controlled experiment with 32 university EFL learners, showing that MAS outperformed single-agent systems (SAS) in oral proficiency gains ($p = 0.049$) and grammatical accuracy ($p = 0.016$).
- Found that MAS led to 26% more practice sessions, 15% longer utterances, and a 70% reduction in repeated grammatical errors.

Intelligent Annotation and Feedback System for English Writing [2] Apr 2025 - Sep 2025
Supervisor: Dr. Zhang SZU

- Developed LLM-IAF (LLM-based Intelligent Annotation and Feedback System), a mobile application for automated English writing assessment for junior high school students.
- Implemented a dual-engine AI workflow combining semantic evaluation and visual grounding to provide immediate visual feedback with error localization on handwritten essays.
- Conducted an experimental study comparing students using LLM-IAF with a control group receiving traditional instruction across four weekly writing tasks.
- Observed significant improvements in writing performance, learning engagement, and writing self-efficacy in the experimental group.
- Demonstrated strong agreement between AI and teacher scoring, and students reported high satisfaction with the system's usefulness and ease of use.

ADDITIONAL RESEARCH

Reasoning for Table Manipulation [3] Mar 2025 - Jan 2026
Supervisors: Prof. Yang, Dr. Tan Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

- Developed an end-to-end LLM framework for table manipulation through structured reasoning.
- Constructed a benchmark covering five core tasks: table splitting and merging, wide-to-long conversion, semi-structured field parsing, and row/column generation.
- Completed two-stage training with supervised fine-tuning on reasoning traces followed by GRPO optimization, achieving state-of-the-art performance among 7B-scale table-specific models.
- Analyzed structural integrity challenges and found that column-level accuracy often exceeded row-level accuracy because of sensitivity to missing fields.

Differential Privacy Image Generation Mar 2024 - Apr 2025
Supervisor: Dr. Zhang SZU

- Developed a differential privacy framework for image generation to reduce gradient clipping bias and improve training stability.
- Introduced reconstruction loss and noise injection during generator upsampling to improve data utility and image diversity.
- Designed a multi-component training pipeline integrating a generator, discriminators, a classifier, and an encoder with gradient sanitization mechanisms.
- Achieved state-of-the-art performance on MNIST and Fashion-MNIST, surpassing baseline methods in Inception Score, Frechet Inception Distance, and downstream classification accuracy.

PUBLICATIONS

[1] Jun Zhang, **Qiwei Ma**, Yu Zhang, and Xiaoming Cao. *Multi-Agent vs. Single-Agent AI for EFL Speaking Practice: A Controlled Experiment with Hybrid Input, Contextual Dialogue, and Proficiency-Adaptive Feedback. Educational Technology & Society (ET&S)*. 2025, (Accepted).

[2] Jun Zhang, **Qiwei Ma**, Yu Zhang, and Xiaoming Cao. *LLM-Based Intelligent Annotation and Feedback: System Design and Empirical Evaluation for English Writing Instruction.* (Expected submission in January 2026).

[3] **Qiwei Ma**, Bo Li, Hengyang Mu, Chuancheng Zhao, Hong Li, Kai Yan, Qingwen Li, Jun Zhang, Minghuan Tan, and Min Yang. *Advancing Large Language Models with Reasoning for Dynamic and Hierarchical Real-World Table Manipulation.* (Expected submission in January 2026).

EXPERIENCE

Research Assistant Apr 2026 - Apr 2027
The Education University of Hong Kong Hong Kong, China

- Research focus: robotics, embodied intelligence, and AI for education.

Research Intern May 2025 - Jan 2026
Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences China

- Worked on table reasoning and manipulation with large language models.